

Acclimator: Jointly Managing Electrical and Thermal Energy in Environmentally-powered Computer Systems

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Environmentally-powered computer systems operate on renewable energy harvested from their environment, such as solar or wind, and stored in batteries. While harvesting environmental energy has long been necessary for small-scale embedded systems without access to external power sources, it is also increasingly important in designing sustainable larger-scale systems for edge applications. Importantly, environmentally-powered systems may be deployed in many different climates that subject them to a wide range of ambient temperatures, which can affect their operation in numerous and significant ways. Thus, these systems must consider, not only the electrical energy, but also the thermal energy available in the environment in both their design and operation. Unfortunately, prior work generally ignores the impact of thermal effects on environmentally-powered systems, and instead implicitly assumes ideal temperatures. To address the problem, we present *Acclimator*, a tool and runtime for designing and operating environmentally-powered computer systems. In evaluating *Acclimator*, we enumerate, quantify, and model the thermal effects that impact these systems using a small-scale prototype and programmable incubator. We then leverage our models to show how considering these thermal effects in designing and operating environmentally-powered computer systems can improve the energy-efficiency, performance, and availability for a wide range of applications.

1 INTRODUCTION

Environmentally-powered computer systems operate on renewable energy harvested from their environment, such as solar or wind, and stored in batteries. While harvesting environmental energy has long been necessary for small-scale embedded systems without external power sources [1, 26], it is also increasingly important in designing sustainable larger-scale systems for zero-carbon and edge applications. For example, there has been a recent emphasis on designing zero-carbon edge-cloud infrastructure powered by renewable energy to mitigate climate change [2, 6]. Since these systems' power is intermittent and limited by their environment, they must carefully regulate their energy usage over time to match their supply. Thus, there has been substantial prior work on designing environmentally-powered systems that dynamically adapt their energy usage to enable perpetual operation, mostly for small-scale energy-harvesting sensor systems [16, 23, 27]. Importantly, environmentally-powered systems may be deployed in different climates that subject them to a wide range of ambient temperatures, which affect their operation in numerous and significant ways.

In particular, changes in temperature can alter the energy-efficiency of processors, batteries, and fans in significant, but different, ways. For example, battery charging and discharging becomes much less energy-efficient as temperature decreases, and may shutdown if the temperature decreases or increases too much. These changes to energy-efficiency due to climate and temperature can significantly decrease these systems' energy-efficiency, performance, and availability. Prior work has generally ignored these effects, and often implicitly assumes an ideal temperature range, e.g. 20-25°C, even though most locations do not experience ideal temperatures year-round. For example, in the midwest and northeast U.S., temperatures can range anywhere between -30°C (-22°F) to 40°C (104°F) over a year. The lack of consideration about thermal effects is one reason

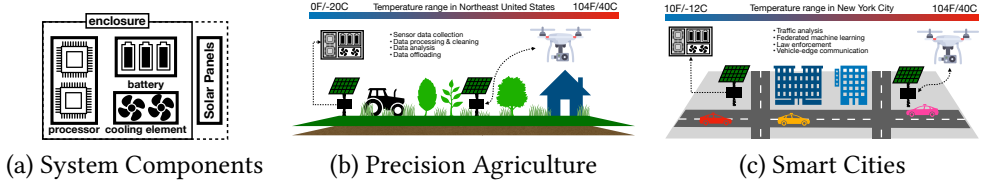


Fig. 1. Environmentally-powered systems typically consist of processors powered by solar panels + battery backups and include a cooling element (a). Common applications include precision agriculture (b) and smart cities (c). In both cases, they are exposed to extreme and widely-varying environmental conditions. A popular precision agriculture application is Farmbeats [26]. Such systems are also being used in smart city initiatives that often include networks of security cameras.

that reported uptimes for environmentally-powered systems such as FarmBeats are often low, e.g., <30 days [17]. Our key insight is that, to optimize performance, energy-efficiency, and availability, environmentally-powered computer systems must jointly consider and manage both the electrical and thermal energy in the environment as part of their design and operation. While much prior work has examined adapting system operation to match variations in available electrical power, e.g., from solar or wind, in both small-scale sensing systems [16, 23, 27] and large-scale cloud systems [11, 12, 18, 22], it has not addressed the significant impact of climate on the design and operation of these systems.

To address the problem, we enumerate, quantify, and model the numerous thermal effects that impact the design and operation of environmentally-powered systems. While the temperature responses of individual components, e.g., processors, batteries, fans, etc., are well-known, optimizing the performance, energy-efficiency, and availability of these systems requires exploiting relationships between these components and their environment. For example, at low temperatures, environmentally-powered systems can leverage some of the thermal energy generated by their processors to heat their battery, which can significantly increase the energy-efficiency of both. Of course, these systems must also efficiently dissipate their heat at high temperatures, e.g., using a fan, to prevent processors and batteries from over-heating and becoming unavailable.

To this end, we design a simple thermodynamic model for environmentally-powered systems by combining well-known physical models of heat transfer, batteries, processors, and fans. Importantly, our model captures *thermal feedback loops* between components that affect the system’s operation, such as how scavenging a system’s waste heat warms its battery, increasing its energy-efficiency, and enabling more computation. We empirically validate our model using a small-scale prototype and programmable incubator that can precisely regulate the temperature between -30°C and 40°C . We then leverage our thermal model to show how considering thermal effects in both designing and operating environmentally-powered computer systems can improve their performance, energy-efficiency, and availability. Specifically, our model and analysis quantifies the effect of a system’s utilization, power draw, enclosure insulation, and ambient temperature on its energy-efficiency, i.e., computational work done using a fully charged battery. We also highlight the tradeoff between an enclosure’s heat transfer coefficient and its energy-efficiency: better insulation, i.e., a lower heat transfer coefficient, increases energy-efficiency when cold by more productively using waste heat, but decreases it when hot by requiring additional energy to power a fan that dissipate heats to prevent battery and processor over-heating. Our work differs from prior work on optimizing the cooling infrastructure of data centers powered by the electric grid, as that work mostly focuses on the efficient movement of heat from *within* the facility to outside it, and does not exhibit the feedback loop between computation and batteries present in environmentally-powered systems.

Accounting for thermal effects is most necessary for climates that experience variable temperatures over the year, including much of the United States, Europe, and other locations at higher

latitudes. These climates require systems to efficiently scavenge heat in the winter, or at night, and dissipate heat in the summer, or during the day, to optimize their operation. By contrast, systems at lower latitudes closer to the equator generally have a more consistent climate—both over the year and day-to-night—which may require less adaptation. As we show, the thermal effects of environmentally-powered systems alters some common assumptions about energy-efficiency. For example, since most systems are not energy-proportional [4], their most energy-efficient operating point (at a fixed frequency and voltage) is generally 100% utilization, since this amortizes their baseload energy across the most computation. However, *this is not true for environmentally-powered systems that use batteries, but is instead dependent on temperature*. At higher temperatures, high utilization is less energy-efficient due to a higher current draw, which reduces battery energy-efficiency, while, at lower temperatures, high utilization is more energy-efficient, since the heat generated by the system warms the battery, which increases its energy-efficiency and offsets the effect of high current.

Our work demonstrates that managing and adapting to variable thermal energy is as important as electrical energy in environmentally-powered systems, and that they are dependent on each other. Currently, thermal management is mostly an after-thought for these systems with most implicitly designed for ideal-to-higher temperatures, often with little insulation that reduces the need for active cooling as temperatures rise, but wastes much of the heat these systems produce as temperatures drop. There is little understanding, and no explicit modeling, of how temperature affects these systems. Our work is an important step towards better understanding how the temperature effects of individual components manifest at the system-level.

Our hypothesis is that optimizing environmentally-powered systems requires jointly managing the electrical and thermal energy in their environment as part of their design and operation. In evaluating our hypothesis, we make the following contributions.

Thermodynamic Model and Validation. We design a simple thermodynamic model for an environmentally-powered computer system that consists of an energy source, e.g., solar panel, enclosure, batteries, and processors that are subjected to some ambient temperature. The model accounts for the effect of heat and processor power on battery capacity, charging, and discharging, the heat emitted by the processor, and the energy consumed by a fan to dissipate heat. We validate our model by enumerating, isolating, and empirically quantifying all of the thermodynamic effects that impact the system’s operation, and how they relate to each other. Our empirical analysis demonstrates the impact of each effect on system operation under different ambient temperatures.

Acclimator Design. We present *Acclimator*, a tool and runtime for designing and operating environmentally-powered computer systems. Acclimator uses our thermodynamic model above to determine offline the optimal system design that satisfies user-specified performance objectives, e.g., for performance, energy-efficiency, and availability, based on a location’s historical weather data. Acclimator then dynamically controls scheduling and cooling system at runtime to optimize operation as weather conditions change.

Implementation and Evaluation. We implement a small-scale prototype and programmable incubator to empirically analyze and validate our model. We also implement a simulator based on our model to enable long-term experimentation. We evaluate the importance of thermodynamic design by showing how different system designs and operating points affect system performance, energy-efficiency, and availability. Finally, we show how Acclimator improves the performance of two case study applications—for precision agriculture and federated learning at the edge.

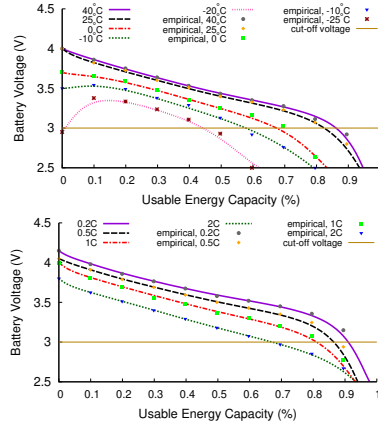
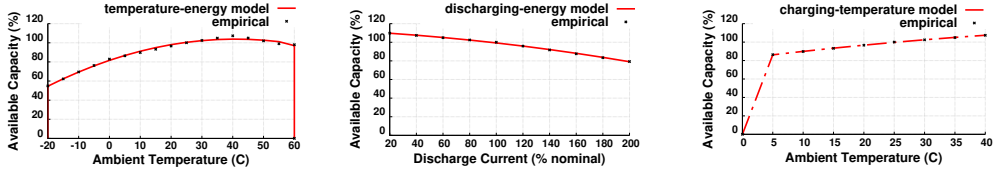


Fig. 2. Usable battery capacity varies with temperature (top) and discharge current (bottom).



(a) Temperature-Energy Effect (b) Discharge-Energy Effect (c) Temperature-Charging Effect

Fig. 3. Computational energy-efficiency as a function of ambient temperature (a), discharge current (b), and available battery capacity for charging at various temperature (c). For (a) and (b), the black points show the experimental data and the red curves show the polynomial fit as the model used by the simulator.

2 BACKGROUND

We briefly summarize the different thermodynamic effects exhibited by batteries, processors, and fans, and how they alter the energy-efficiency of each. We model these effects in the next section, and then validate and quantify them.

2.1 Environmentally-powered Systems

Environmentally-powered computer systems operate on renewable energy harvested from their environment, such as solar or wind, and stored in batteries. Figure 1(a) shows the typical components of these systems, which include solar panels, processors, batteries, and a cooling element, such as a fan or pump. Figure 1(b) & (c) show two of the applications that leverage environmentally-powered systems. Precision agriculture applications deploy these systems to gather data from small-scale embedded systems that use low-power sensors to monitor environmental conditions, such as soil moisture, humidity, and temperature. These applications may also gather and process imagery from cameras. Similarly, there are a wide range of smart city applications, such as traffic monitoring, vehicle-to-edge communication, and crime detection, that analyze and process the data near the edge collected by sensors and cameras.

2.2 Batteries

The energy stored by batteries is related to their temperature and discharging/charging current. We discuss these relationships below.

2.2.1 Temperature-Energy Effect. The usable energy capacity of lithium batteries significantly decreases with temperature. Figure 2(top) shows curves from our prototype's battery datasheet, where the points represent experiments we have run to empirically validate the datasheet using our

programmable incubator. The graph shows that the battery's usable energy capacity, as a percentage of its charged capacity, drops by over 50% at the 3V cut-off voltage when the temperature drops from 25°C to -20°C. This "wasted" energy is consumed as heat by the battery to catalyze its chemical reaction that produces electricity. We call this the *temperature-energy* effect. In addition, discharging at low temperatures can also damage batteries and reduce their cycle-lifetime. In general, lithium batteries should not be discharged below -20°. Likewise, as temperatures increase, the lithium batteries' self-discharge rate also increases, which reduces the energy available for discharge, although not by as much as a decrease in temperature. Lithium batteries generally cannot be discharged safely at >60°, and thus at that temperature their available capacity drops to 0.

Figure 3(a) plots the temperature-energy effect in our prototype with ambient temperature on the x-axis and available battery capacity on the y-axis. The points represent experiments using our prototype, while the continuous line represents our model's prediction, which closely matches the data. In this case, we set the battery to 100% capacity at 25°C. As shown, the available capacity drops significantly as the temperature decreases, with only 80% of the energy available at 0°C and 50% available near -20°C. Temperatures between 0°C and -20°C are common over winter in much of the U.S., Europe, and other high latitude locations. As we show, the heat generated by processors can be leveraged by environmentally-powered systems to raise their internal enclosure temperature and extract more energy from their batteries. The plot also shows how our prototype's charge controller automatically shuts down the battery once it reaches 60°C for safety. While ambient temperatures on earth generally do not reach 60°C, they can reach this high within an insulated enclosure if processors generate heat faster than the system can dissipate it, i.e., via conduction or convection.

2.2.2 Discharge-Energy Effect. The *discharge-energy effect* refers to the decrease in available energy that occurs when discharging at higher rates. Figure 2(bottom) shows curves from our prototype's battery datasheet, where the points represent experiments we have run for empirical validation. The graph shows that, at the 3V cut-off voltage, the battery's usable energy decreases by 25% when increasing the current from a C-rate of 0.2 to 2, where a C-rate of N represents the discharge current required to fully discharge the battery in $1/N$ hours. Due to this effect, if processors execute at 100% utilization, they are able to draw less energy in total from the battery than if they operate at a lower utilization, since utilization is roughly linear with current draw (when processors operate at a fixed voltage). Thus, the slower processors execute, the more energy they can extract from the batteries, and the more overall computation they can perform.

Figure 3(b) quantifies the discharge-energy effect on our prototype where we plot the discharge current (which is linear with utilization) on the x-axis and available energy capacity on the y-axis. Again, the points represent experiments with our prototype, while the continuous line represents our model's prediction, which closely matches the empirical data. Here, we normalize the experiment to some discharge current (equivalent to 50% utilization), which we set at 100%, and then set the available capacity at 100% for that discharge current. The experiment shows that the relationship is nearly linear, as we slow down the system (by reducing its utilization and current draw), we are actually able to extract more than our nominal available energy, and as we speed up the system (by increasing its utilization and current draw), we draw less energy (in this case, only 80% capacity is available when operating at 100% utilization compared to 50% utilization). Note that the discharge-energy effect counteracts the temperature-energy effect above: running the processor at high utilization generates more heat, which can warm the battery and increase its available energy, but it also increases the discharge current, which reduces the available energy. Thus, the most energy-efficient operating point at any given time is challenging to determine.

2.2.3 Temperature-Charging Effect. The *temperature-charging effect* refers to the relationship between temperature and the rate at which batteries can charge. While lithium batteries can safely discharge down to -20°C , their maximum charging rate decreases with temperature and is not possible below 0°C . Thus, *low temperatures prevent storing energy and make using energy much less efficient*. Lithium batteries can also easily overheat at high temperatures, since their chemical reaction generates additional heat that increases their internal temperature beyond the ambient temperature. Charge controllers generally prevent charging/discharging when the internal temperature rises too high (above $\sim 60^{\circ}\text{C}$). Thus, *high temperatures can also prevent storing and using energy*. Figure 3(c) shows how the charging capacity of the battery changes with temperature. In this case, 100% represents the maximum charging capacity at 25°C . As the graph shows, the charging rate decreases rapidly from 5°C to 0°C , where it falls to 0%. At 5°C the capacity is $\sim 80\%$ and then increases roughly linearly with temperature. As expected, higher temperatures enable the system to charge the battery at faster rates. As above, the points represent experiments using our prototype, while the continuous line represents our model's prediction, which closely matches.

2.3 Processors

Since processors do no mechanical work, their power is converted to heat, which must be dissipated to prevent them from overheating due to thermal runaway. A computer system's energy-efficiency is a function of temperature if it leverages outside air (or water) for cooling, since lower ambient temperatures require using less additional energy to actively cool the processor. Such "free cooling" is often used by cloud data centers [13]. Thus, unlike batteries, processors are *more* energy-efficient at low temperatures, since they do not have to consume additional power to dissipate heat. Also unlike batteries, computer systems (at a fixed frequency and voltage) are more energy-efficient, in joules per computation, at higher power, since they are generally not energy-proportional and a higher power (and utilization) amortizes their baseload over more computation.¹ Of course, an ideal energy-proportional system, by definition, has the same energy-efficiency at any utilization. As we discuss, the heat generated by processors can be recycled by environmentally-powered systems to optimize the efficiency of the battery based on the various effects above. For example, scheduling more workload when temperatures are low can improve system performance by generating heat that improves battery efficiency. We call this the *scheduling-performance* effect.

2.4 Cooling System

Fans (or pumps) are used to dissipate heat in environmentally-powered systems. The intensity with which a variable speed fan must run to maintain a certain temperature is a function of the heat dissipated by the processor, the conductivity of the system enclosure's insulation, and the external temperature. Thus, there is an *insulation-fan effect* that impacts system enclosure design: the thicker the enclosure's insulation, the better the performance in low temperatures, but the more the fan must run in high temperatures, and vice versa. In addition, as we discuss, fan (and pump) energy usage is a cubic function of the amount of air (or water) it moves, and thus the heat it dissipates, which impacts its effectiveness.

The *insulation-fan effect* refers to the tradeoff between having thicker insulation to retain heat during low temperatures at the cost of consuming more energy via fan to dissipate heat at high temperatures. Figure 4 quantifies the insulation-fan effect for our prototype at 25°C for different levels of insulation on the x-axis. The y-axis shows the power required by the fan to maintain 25°C as the enclosure's insulation increases when operating the system at 50% utilization. As expected, when the insulation is thin, there is almost no need for the fan, and it consumes little power.

¹This is not necessarily true when using DVFS to reduce the frequency and voltage, which super-linearly increases processor energy-efficiency.

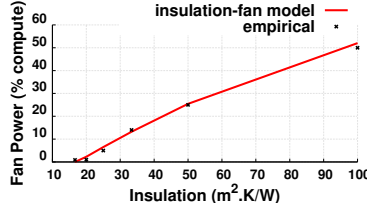


Fig. 4. Power consumed by the fan as a function of insulation. The purple line plots the model used by the simulator based on a polynomial fit of the experimental data.

However, as we increase the insulation’s thickness, the enclosure retains more heat, which requires consuming more power to dissipate that heat by the fan. Of course, the energy used by the fan is energy that does not go towards productive computation. One option at these higher insulation levels, instead of running a fan, is to operate at a lower utilization to generate less heat, which reduces the need to consume energy by the fan to dissipate heat. Thus, as with the discharge-energy effect, operating slower, at a lower utilization, enables more energy to go towards productive computation. That is, active heat dissipation using fans enables environmentally-powered systems to operate at higher workload intensities than they otherwise could, but at the cost of lower energy-efficiency. Also, as above, the points in the graph represent experimental data from our prototype, while the line represents our model’s prediction, which closely matches.

3 ACCLIMATOR DESIGN

In this section, we present the design of *Acclimator*, a tool and runtime for designing and operating environmentally-powered systems. We first provide an overview of different components and then go into the details of each component.

3.1 Acclimator Overview

Acclimator’s goal is to enable the energy-efficient, reliable, and performant design and operation of environmentally-powered embedded systems. *Acclimator* consists of three components: a thermodynamic model, *acclimator* design module, and an *acclimator* operation module. The overall architecture of the *acclimator* and the relationship between its different components is shown in Figure 5.

The heart of the *Acclimator* is a thermodynamic model that captures the various effects that impact the performance of environmentally-powered systems. We describe and validate the thermodynamic model in Section 3.2. The design tool, for a given location and user-specified design objectives, outputs a range of enclosure specifications that enables the user to achieve the desired performance objectives at the specified location based on its historical climate. This is an offline, one-time process and is done before the system deployment. We provide more details about this process in Section 3.3. The operation module operates at runtime and schedules the energy usage of the system in a way that achieves the desired metric in the presence of a dynamic workload. We detail our online scheduling approach in Section 3.4.

3.2 Thermodynamic Model

To better understand the effect of temperature on the operation of an environmentally-powered system, we develop a simple physical thermodynamic model that estimates a system enclosure’s change in temperature over some time interval Δt . Figure 6 illustrates our model and its key parameters. The model assumes processors and batteries reside within an enclosure of some size. The processors, batteries, and the air within the enclosure each have an associated temperature (T), mass (m), and thermal capacity (C), which is the heat required to change the temperature of

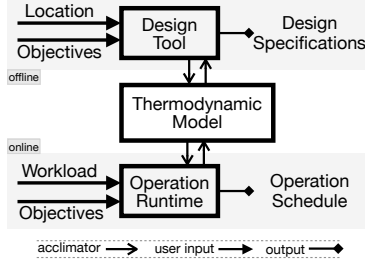


Fig. 5. An overview of *Acclimator* including its key modules, input requirements, and output parameters.

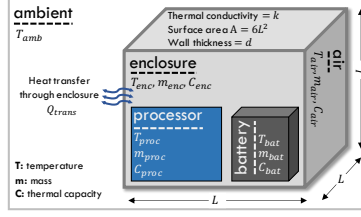


Fig. 6. A simple thermodynamic model of environmentally-powered computer systems.

the mass, and is in units of joules per degree K. We also assume the ambient temperature outside (T_{amb}) is unaffected by any heat transfer with the enclosure.

The enclosure provides some insulation from the ambient environment based on its heat transfer coefficient U , which is an empirically derived constant that dictates the heat transfer rate ($\dot{Q}_{trans}$) in joules per unit time between the enclosure and its external ambient environment. The overall heat transfer coefficient U is a combination of the internal convection inside the enclosure, conduction through the enclosure walls, and external convection away from the enclosure. We can calculate U as a set of thermal resistors connected in series, as shown below.

$$\frac{1}{U} = \frac{1}{h_i} + \frac{d}{k} + \frac{1}{h_o} \quad (1)$$

In the equation above, h_i is the internal convection coefficient, k is the thermal conductivity, d is the thickness of the enclosure, and h_o is the outer convection coefficient. The heat transfer rate ($\dot{Q}_{trans}$) below between the enclosure and its external environment is directly proportional to the heat transfer coefficient (U), temperature difference ($\Delta T(t)$) between inside and outside the enclosure, and the heat transfer area (A), as shown below.

$$\dot{Q}_{trans} = U \times A \times \Delta T(t) = U \times 6L^2 \times (T_{amb} - T_{enc}(t)) \quad (2)$$

For simplicity, our model assumes the enclosure is a cube with side length L , so its surface area $A = 6L^2$. As shown above, the heat transfer rate is a function of this surface area A . At any time t , $\Delta T(t) = T_{amb} - T_{enc}(t)$, representing the difference between the temperature inside and outside the enclosure. Thus, a positive $\Delta T(t)$ represents heat flowing into the enclosure, and a negative $\Delta T(t)$ represents heat flowing out of it. Given the equation above, we can compute the total heat transfer (in joules) over some time interval Δt by simply integrating $\dot{Q}_{trans}$ over time, as shown below.

$$Q_{trans} = \int_t^{t+\Delta t} \dot{Q}_{trans} dt \quad (3)$$

Equation 3 enables us to compute the total heat energy transferred between the enclosure and outside environment. However, some of this heat energy is absorbed by the mass within the enclosure, including the processors, batteries, and air, and thus does not contribute to raising the enclosure's temperature (T_{enc}). This heat energy is a function of the enclosure's heat capacity (C_{enc}),

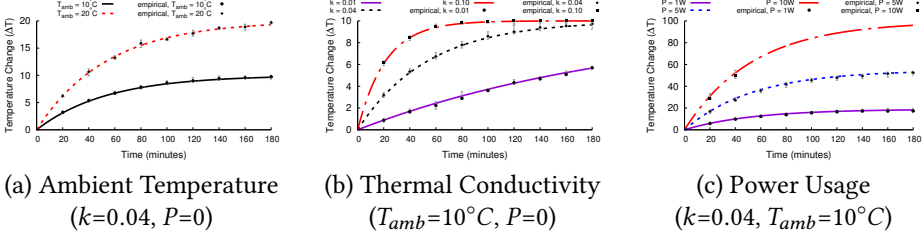


Fig. 7. The change in the enclosure's temperature T_{enc} over time is a function of the (a) ambient temperature T_{amb} , (b) the enclosure's thermal conductivity k , and (c) the processor's power usage P .

which is computed as the mass-weighted average of the respective heat capacities of the objects within the enclosure, as shown below, where $m_{enc} = m_{air} + m_{bat} + m_{proc}$ or the total mass within the enclosure. Here, we assume the enclosure includes only processors, batteries, and ambient air.

$$C_{enc} = \frac{1}{m_{enc}} \times (m_{air} \cdot C_{air} + m_{bat} \cdot C_{bat} + m_{proc} \cdot C_{proc})$$

We can empirically measure the battery and computing platform's mass (m) and heat capacity (C) using a scale and calorimeter, respectively. We cannot directly weigh the air mass, but can derive it using simple models. In particular, the mass of air in the enclosure is a product of its volume and the air's density ρ_{air} , which is directly proportional to the atmospheric pressure, and inversely proportional to the temperature (T_{air}) as well as its specific gas constant ($R_{specific}$), as shown below.

$$\rho_{air} = \frac{p}{R_{specific} \times T_{air}} \quad (4)$$

For dry air on earth $R_{specific} = 287.058 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$. We assume the enclosure is closed when $p = 1$ atmosphere and $T_{air} = 25^\circ\text{C}$. Note that, based on the ideal gas law, even when the temperature inside the enclosure changes, the ratio of its pressure p to its temperature T_{air} , and thus its air density, remains constant. As a result, in this case, the density of air $\rho_{air} = 1.1839 \text{ kg/m}^3$, which results in an air mass $m_{air} = 1.1839 \times L^3$. The heat capacity of air (C_{air}) at earth's surface under these conditions is also a constant and equal to 717 joules per kilogram per degree Kelvin (K).

Given all this, we can compute the enclosure's heat capacity C_{enc} above. If the enclosure generates no internal heat, then its temperature will eventually reach an equilibrium temperature equal to the temperature T_{amb} of the ambient environment. To reach equilibrium, the total amount of heat Q_{trans} the enclosure will absorb or release is the product of its total mass m_{enc} , heat capacity C_{enc} , and change in temperature, as shown below.

$$Q_{trans} = m_{enc} \times C_{enc} \times (T_{amb} - T_{enc}) \quad (5)$$

While the equation above represents the heat transferred to reach the equilibrium temperature, the same basic equation also dictates the heat transferred over any arbitrary time interval Δt based on the change in temperature at the time interval's start and end, as shown below.

$$Q_{trans} = m_{enc} \times C_{enc} \times (T_{enc}(t + \Delta t) - T_{enc}(t)) \quad (6)$$

Notice that we have computed the total heat transfer Q_{trans} over a time interval in both Equation 3 and Equation 6. Setting these equations equal to each other yields our model, which predicts the temperature within the enclosure after some time interval Δt given a starting temperature $T_{enc}(t)$, the enclosure's mass (m_{enc}) and heat capacity (C_{enc}), as well as its thermal conductivity (k), surface area (A), ambient temperature (T_{amb}), and depth (d).

$$m_{enc} \times C_{enc} \times (T_{enc}(t + \Delta t) - T_{enc}(t)) = \int_t^{t+\Delta t} \hat{Q}_{trans} dt$$

$$T_{enc}(t + \Delta t) = T_{enc}(t) + \frac{1}{m_{enc} C_{enc}} \int_t^{t+\Delta t} \hat{Q}_{trans} dt \quad (7)$$

To this point, our model assumes the processor generates no heat. In practice, however, the power drawn by the processor is converted to heat, which our model assumes is uniformly distributed throughout the enclosure. At this point, we assume there are no mechanical components, such as fans, to dissipate this heat. We discuss modeling for heat dissipation using a fan below. Thus, we extend our model above to account for the processors' power draw by assuming it is entirely converted to heat. We can account for this heat energy in the model above by simply adding it to the heat transferred with the environment, as shown below.

$$T_{enc}(t + \Delta t) = T_{enc}(t) + \frac{1}{m_{enc}C_{enc}} \int_t^{t+\Delta t} \hat{Q}_{trans} + V \cdot I \, dt$$

Here, V is the platform's voltage, I is its current draw, and $V \cdot I$ is its overall power draw. The model above simply observes that power translates directly to heat within the enclosure and thus augments any other heat transfer.

Our model above captures an enclosure's temperature change over time based on its physical characteristics, the ambient temperature, and the processor's power usage. Unfortunately, there are no good physical models that capture the effect of temperature and power draw on usable battery capacity. Thus, we use the empirical models from the battery's datasheet in Figure 2, which we experimentally validated. We next discuss extending our model to include active cooling using a fan.

3.2.1 Heat Dissipation Using a Fan. Processors may dissipate heat faster than an enclosure can transfer it to the external environment based on its conductivity, which causes the temperature to rise to a point where neither the battery nor processor can function. In this case, fans or pumps may be necessary to increase the rate of heat dissipation within the enclosure using convection. While our model below focuses on fans, which transfer heat by moving air, the same basic approach applies to pumps, which transfer heat by moving a liquid. The selection of a fan depends on the specifications of the enclosure and the maximum rate of heat dissipation required. A fan moves the air that absorbs the heat from inside the box and then dissipates it to the external environment. The amount of energy dissipated (Q_{diss}) depends on the mass of the moving air (m_{air}), the specific heat of the moving air (C_{air}), and the temperature change of the moving air (ΔT_{air}).

$$Q_{diss} = m_{air} \times C_{air} \times \Delta T_{air}$$

The mass of the moving air can be calculated from the volume of air (V_{air}) being moved and the density of the moving air (ρ_{air}).

$$Q_{diss} = (V_{air} \times \rho_{air}) \times C_{air} \times \Delta T_{air}$$

We divide both sides of the equation to get the power needed to dissipate at each instance of time (t).

$$\frac{Q_{diss}}{t} = \left(\frac{V_{air}}{t} \times \rho_{air} \right) \times C_{air} \times \Delta T_{air}$$

The air volume over time is the air flow rate, which we term as AF_{air} . We term the power dissipation as P_{diss} . We arrange the equation to get the airflow required for a given power dissipation.

$$AF_{air} = \frac{P_{diss}}{\rho_{air} \times C_{air} \times \Delta T_{air}}$$

The value of C_{air} is $1kJ \cdot kg^{-1} \cdot C^{-1}$ and the density of air at $20^\circ C$ is $\rho_{air} = 1.20kg/m^3$. We can use this equation to either find the heat dissipation rate for a given flow rate or the heat flow required to achieve the desired heat dissipation rate.

Since the equipment inside the enclosure will exhibit resistance to the air flow, air needs to be delivered at a certain pressure that can overcome the resistance. However, the amount of pressure required is highly dependent on the design and physical characteristics of the product to be cooled,

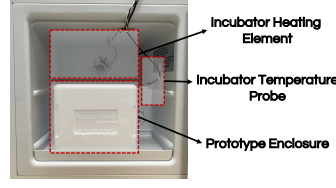


Fig. 8. Small-scale prototype inside the incubator.

and must be determined either experimentally using anemometers and manometers to measure the air speed and pressure, respectively, or using different computer-aided design (CAD) software to design and calculate airflow characteristics. Either method will yield system pressure requirements that increase with the air flow. The pressure exerted by the fan reduces as its air flow increases, and delivers the highest air flow when the back pressure is lowest. The intersection of the two curves is the operating point of the fan for the given system. We assume our model uses a fan that has the required airflow at its operating point.

The power consumption of a variable speed fan has a cubic relationship with the change in the airflow. That is, if the airflow of the fan doubles, its power consumption increase by 8×. This relationship is shown in the equation below.

$$P_j = P_i \times (AF_j / AF_i)^3$$

Here, P_i and AF_i are the initial power consumption and airflow while P_j and AF_j are the final power consumption and airflow, respectively. Note that using the fan to dissipate heat reduces the energy available for doing productive computation.

3.2.2 Implementation and Model Validation. To validate our model and experiment with thermodynamic design, we built a programmable incubator by connecting a mini-freezer and incandescent light bulb (as a heat source) to programmable relays controlled by a Raspberry Pi (Figure 8). Our incubator programmatically controls temperature between -30°C and 40°C with an error of $\pm 2^\circ\text{C}$. We use the Nvidia Jetson Nano as our computing platform for validation. The Nano has a baseload power of $\sim 1\text{W}$, and a maximum power of 10W . We use Panasonic NCR18650B lithium-ion batteries rated for 3.2 Amp-hours (Ah) at 3.6V. We use a boost converter to build a 4Ah, 5V battery bank for the Nano with 20Wh maximum energy capacity. Figure 2 from §2 shows our battery's response to temperature and discharge current. Our enclosure uses Expanded Polystyrene (EPS) foam, which has a thermal conductivity k of $0.04\text{Wm}^{-1}\text{K}^{-1}$ [19]. We use 1.25in thickness as our baseline for experiments. We vary k by changing its thickness. For example, halving the thickness increases k by $2\times$ to $0.08\text{Wm}^{-1}\text{K}^{-1}$.

Three parameters affect how the enclosure's temperature T_{enc} changes over time: the difference with the ambient temperature T_{amb} , the enclosure's thermal conductivity k , and the processor's power usage P . Figure 7 shows the complex non-linear effect of each on the change in T_{enc} . Our baseline is $T_{amb}=10^\circ\text{C}$, $k=0.04$, and $P=0\text{W}$. Figure 7(a) then varies T_{amb} without changing k or P , and shows that a higher ambient temperature causes the enclosure's temperature to rise more quickly. Similarly, Figure 7(b) varies k , and shows that a higher thermal conductivity, i.e., more insulation, also causes the temperature to rise more quickly. Finally, Figure 7(c) varies the processor's power usage, and shows how the resulting heat dramatically increases the temperature up to 100°C at full utilization (10W), which is well beyond the 10°C ambient temperature. Figure 7 also validates our model: the curves represent our model, while datapoints represent the mean temperature change across five experiments, where error bars represent the maximum and minimum. As shown, the data closely matches our model.

3.3 Acclimator: The Design Tool

The goal of this tool is to determine the configuration range for the system enclosure that allows the system to meet the performance objectives across seasons. In addition to the thermodynamic model, the tool has two key components. The first component is a weather module that provides historical temperature data for the user-specified location. The second component is *Enclosure Specification Estimator* that exhaustively searches the parameter space to find a set of configurations that satisfy system objectives. We next describe how a user specifies system performance goals, additional inputs required, and finally the output of the design tool.

3.3.1 User Specifications. To use the design tool, a user specifies the location of the deployment, system configuration, performance objectives, and operational constraints. We detail each of these inputs next and discuss how a user can describe each of these parameters before deploying the system.

Location: The user specifies the location in latitude and longitude, e.g., (45, -80). The location is used to retrieve historical temperature data, which is a pre-requisite for the thermodynamic model.

System Configurations: The user defines the system configurations in terms of the system's baseline power consumption, its dynamic range, battery size, and the size of the solar panel. These configurations are known to the user for a well-designed system, as they are needed to make sure the system components are sized appropriately. For example, in the case of FarmBeats [26], the size of the battery and solar panel is known. Furthermore, since all of the system components are known, the base power and the dynamic range of power can be determined experimentally.

Performance Objectives: A user defines the performance objectives using three key metrics that quantify the fidelity of a system. The three metrics are *energy-efficiency*, *availability*, and *work rate*. We next define these metrics.

Energy-efficiency is defined as the percentage of available energy in the battery that is extracted and used for computation.

$$\text{Energy Efficiency} = 100 \times \frac{\text{Energy used for computation}}{\text{Energy stored in the battery}}$$

A value of 100% means that all the energy extracted from the battery was used for the computation, while a value of 50% means that either only 50% of the energy could be extracted and used for the computation. In this case, the other 50% either could not be extracted from the battery or was used by the cooling system. A higher value of energy efficiency is better.

Availability is the percentage of time the system was up and has enough power to operate at or above a threshold utilization.

$$\text{Availability} = 100 \times \frac{\text{System up time}}{\text{Total experiment time}}$$

The value of *availability* value ranges between 0 and 100. A higher value is better for a given operating point.

Finally, the *work-rate* is defined as the amount of work done (computation) per unit time. Its value can be in the $(0, \infty)$ range. The higher value of work rate is better.

To define performance objectives, the user specifies the order of importance for the metrics. For example, a user can specify availability as a primary metric, followed by work rate and energy-efficiency. Our design tool will select the enclosure and cooling element specification, such that the primary metric is ensured while the other objectives are maximized in the order of priority.

Operational Constraints: This is specified as the minimum power level required by the system at all times. For example, to ensure a certain data acquisition rate, a user deploying a FarmBeats-like system may specify a power level of 30% (relative to the maximum system power) as a constraint that must be met at all times.

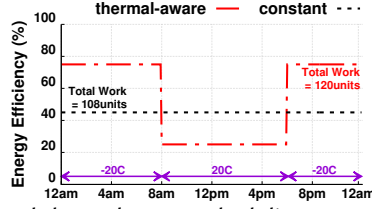


Fig. 9. Work done using a naive and thermal-aware scheduling approach that exploits the feedback loop between electrical and thermal energy.

3.3.2 Enclosure Specification Estimator. Our design tool takes an empirical approach to finding the right specifications for the enclosure and the cooling element. It starts with three values for the insulation level (or thermal conductivity, k) of the enclosure: low (e.g., styrofoam, $k = 0.04W/m.K$), medium (e.g., Polyvinyl Chloride (PVC) plastic $k = 0.2W/m.K$), and high (glass, $k = 0.8W/m.K$). These values are configurable and correspond to actual insulation materials that can be used for the enclosure. Similarly, it picks an initial value for the cooling capacity, specified in Watts (W).

The tool uses an iterative process to find the optimal enclosure and cooling element specifications. It uses the thermodynamic model and plays out the temperature and solar power profiles for a typical year at the different insulation materials above. Since the exact pattern of the workload is not known, it varies the system’s operational power between the user-specified minimum (e.g., 30%) and 100% power. For each of the configuration combinations (i.e., thermal conductivity, cooling capacity, and operational power), it computes the value of all of the metrics that quantify system performance objectives. It then keeps all the configurations that meet the core metric requirements specified by the user. In the next iteration, it optimizes for the second metric of choice and keeps the configurations that maximize this metric. Finally, the last step chooses the set of configurations that maximizes the last metric.

We use a brute-force approach to find the best set of configurations. Since finding the right set of enclosure specifications is a one time process done before system deployment, the computational-intensive nature of the brute-force search is not a problem. However, other optimization techniques, such as Broyden-Fletcher-Goldfarb-Shanno algorithm [15], can be used. However, optimizing the enclosure specification estimation is outside the scope of this paper.

3.3.3 Module Output. Finally, the *Acclimator* design tool outputs a single configuration, or a set of configurations, that satisfy the desired performance objectives. The user is also informed if no configuration could meet the desired level of performance under the operational constraints provided by the user.

3.4 Acclimator: Operation at Runtime

The *Acclimator* operation at runtime consists of three key components: a forecasting module, a simulator, and a scheduler.

Forecasting Module. The forecasting module is responsible for forecasting the workload, the ambient temperature, and the solar power. The environmentally-powered computer systems we consider are often designed for specific applications, such that their workload patterns are well-known and easy to predict. Thus, we assume an accurate workload forecast is available. Our forecasting module leverages publicly-available temperature forecasts, which are generally accurate, and an open-source solar forecasting tool [5]. We assume the forecasts above are accurate; evaluating the effect of inaccurate forecasts is outside the scope of this paper.

The Simulator. The core component of the simulator module is the thermodynamic model. Given the future workload, temperature profile, and the power generation pattern, the *simulator* simply plays out the future over a specified time period.

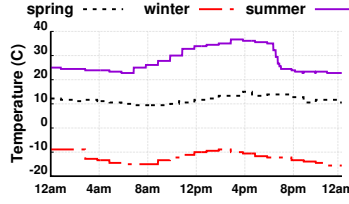


Fig. 10. Temperature profiles of a single location in northeast United States across different seasons: winter (January 15, 2022), spring (April 25, 2021), and summer (June 30, 2021).

The Scheduler. The scheduling module schedules the workload for each hour of the scheduling horizon, while satisfying the operational power constraint (minimum power needed at all times) specified by the operator. To do so, it leverages the key insight that thermal and electrical energies exhibit a feedback loop between them. We use a simple example to demonstrate this effect, which we term the *scheduling-energy effect*. It refers to the relationship between the intensity at which the processors operate at, and the energy available from the battery. Specifically, due to the temperature-energy effect described in Section 2, the warmer the battery, the more energy systems can extract from the battery. Thus, when scheduling workload, systems can extract more energy, and perform more total computation, if they operate at a higher utilization as the temperature decreases, and lower utilizations when the temperature increases. The former maintains a higher temperature, which increases battery efficiency, while the latter generates less heat, which reduces the need to operate a fan, which consumes additional energy to dissipate heat. Figure 9 quantifies this scheduling-energy effect in a scenario where the temperature drops over night but rises during the day. The graph compares operating continuously at $\sim 50\%$ utilization with a scheduling policy that operates at $\sim 25\%$ utilization during the day and $\sim 80\%$ utilization over night. In this case, the latter schedule is able to perform 11% more computation than the former because of the effects above. For this experiment, our computations are simply a benchmark composed of integer operations. The experiment demonstrates that scheduling *when* and *how much* processors dissipate heat can affect a system’s energy-efficiency and the total computation.

The scheduler finds an optimal workload schedule that meets the performance objective outlined by the user. We use a simple iterative approach for finding the right schedule. Our scheduler works backwards from the end of scheduling horizon and schedules workload for each hour such that the energy is extracted from the battery at the highest energy-efficiency. In the first round, it simply uses the default workload pattern and then changes the workload in each slot to make sure the performance objective is met.

4 EVALUATION

In our evaluation, we show the working of *Acclimator* design tool and operation runtime. For the design tool, instead of just presenting the final output of the design tool for our example, we demonstrate how different designs impact a system’s energy-efficiency, performance, and availability for a given operating point. We also show how, for a given design, the operation must adapt to optimize for energy-efficiency, performance, and availability across different climates and seasons. Finally, to demonstrate the efficacy of operation runtime and end-to-end working, we show how *Acclimator* improves the performance of two case study applications—for precision agriculture and federated learning at the edge.

Evaluation Setup. For the *Acclimator* design tool and operation runtime evaluations, we consider a small-scale carbon-free edge datacenter similar to those considered in recent work [2, 6]. The choice of edge datacenter also demonstrates that our approach is applicable at various scales. We consider the enclosure, in this case a small room, as a cube with 8ft sides. The size of the battery

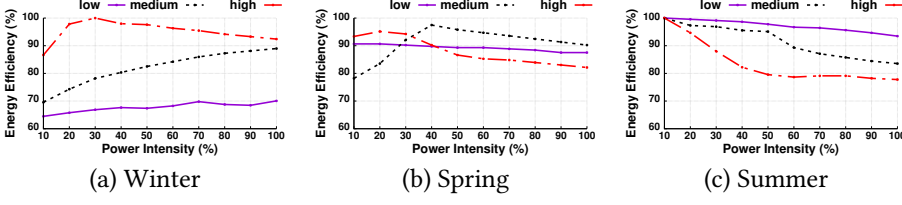


Fig. 11. Energy-efficiency across three seasons: (a) winter, (b) spring, and (c) summer. For each season, we evaluate our three designs that are winter-optimal (red line, long dash), spring-optimal (black line, small dash), and summer-optimal (purple line, solid) over various operating points represented by power intensity of x-axis.

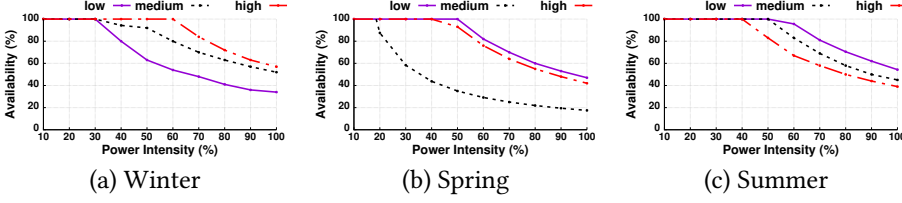


Fig. 12. Availability across three seasons: (a) winter, (b) spring, and (c) summer. For each season, we evaluate our three designs that are winter-optimal (red line, long dash), spring-optimal (black line, small dash), and summer-optimal (purple line, solid) over various operating points represented by power intensity of x-axis.

is 20kWh with a minimum state of charge of 40% or 8kWh. The room houses 8 servers of 250W each with a total demand of 2kW at 100% utilization. The available battery capacity of 12kWh is enough to run all the servers at 50% utilization for 24 hours under ideal conditions. The heat capacity, volume, and mass of the battery is based on a lithium-ion battery datasheet [7].

We demonstrate the effect of the design and operating point on the energy-efficiency, availability, and performance using a site in northeast U.S. This site exhibits significantly different weather patterns across three different seasons: winter, summer, and spring. The temperature profiles for three representative days of these seasons are shown in Figure 10. The temperature varies from 77°F (25°C) to 97°F (36°C) in summer, from 46°F (8°C) to 52°F (11°C) in spring, and 5°F (-15°C) to 16°F (-5°C) in the winter. We configure the heat transfer coefficient of the enclosure walls at three insulation settings: low (2), medium (0.9), and high (0.35). These values are achieved by varying the thickness of insulated brick wall with thermal conductivity of 0.15 W/m·K. Furthermore, all the values fall in the range of a heat transfer coefficients for typical materials.

4.1 Evaluating Design and Operation Tools

To demonstrate the complete landscape of design and operation, we run the *Acclimator* design tool using multiple combinations of performance objectives (energy-efficiency, availability, work rate) and operational power constraints (varied from 10% to 100% with 10% increments). We also decouple the results across three seasons (summer, winter, spring) to demonstrate how the performance trade-offs are impacted by the climate.

4.1.1 Energy-efficiency. Figure 11 shows the energy-efficiency evaluation for different design parameters and operating points across three seasons. The following discussion assumes that the system optimizes for energy-efficiency and discusses the design choices and operating points to achieve that across seasons.

Energy-efficiency vs. Design. The choice of design to optimize for energy-efficiency depends on which season the system optimizes for and how much loss of energy-efficiency it is willing to accept in other seasons. If energy-efficiency in winter is desired for the system, the user should opt for a design that offers the highest protection against ambient weather and best performance in heat scavenging, which we term as winter-optimal (high insulation). This design option gives you

100% energy efficiency in winter and offers the highest energy-efficiency for any operating point (Figure 11(a)). However, its energy-efficiency in other seasons is significantly lower, especially in summer (Figure 11(c)). This is because, in summer, this design has to use significant fan energy to dissipate the heat produced by computation. Similarly, a high heat-transfer coefficient design is the best choice if summer energy-efficiency is desired. As expected, its performance in winter is the worst as high conductivity allows the heat from the computation to escape, preventing it from retaining heat when idle. The spring-optimal design (medium insulation) offers the best performance in spring and fall (Figure 11(b)). Its performance is better than low insulation in winter and high insulation in summer. Therefore, it is the best choice if the energy-efficiency across all seasons is desired.

Key point. *The optimal choice of design depends on the target season/climate. The design can be optimized for any season with an energy-efficiency loss in other seasons. If no particular season is important, medium insulation design is the best choice.*

Energy-efficiency vs. Operating Point. The design of a system for a particular season does not automatically guarantee the best energy-efficiency. The operating point, i.e., utilization, provides another knob that optimizes the energy-efficiency. For example, during winter, the energy-efficiency for the low insulation design is highest at the maximum operating point. This is because, at higher operating points, more heat is generated which keeps the batteries warm. The gain in battery energy-efficiency is enough to offset the negative effect of higher current draw. However, the same design offers the best energy-efficiency at the lowest operating point in spring and summer. This is because, at low operating points, the low insulation is able to dissipate the heat through normal heat transfer. The low operating point not only avoids the use of a fan but also the negative impact of higher discharge currents. This trend is not same for all the design choices. For a high and medium insulation, the best performance is achieved at mid-operating points in their respective seasons. This demonstrates that, given a design, the choice of operating point will vary within a season and across seasons. Figure 11 can also be used as a guide to designing systems. If the system must operate at a certain operating point, you can choose a design that gives the best performance. For example, if the system must always operate at at least 10% power intensity, the high insulation gives the best performance both in winter and summer, and comparable performance for the rest of the year.

Key point. *Given a design, the operating point must adapt to optimize efficiency in each of the seasons. Additionally, the choice of an operating point for a given season dictates the choice of a particular design, which motivates jointly optimizing design and operation.*

4.1.2 Availability. Figure 12 presents the availability evaluation for different design parameters and operating points across three seasons. The following discussion assumes that the system optimizes for availability and discusses the design choices and operating points to achieve that across seasons.

Availability vs. Design. The availability achieved across all the operating points differs for each insulation level. The maximum availability offered by each insulation across all operating points differs significantly across seasons. A high insulation offers a minimum of 60% availability across all operating points as compared to 35% for the low insulation. This is because the energy-efficiency of the two designs varies significantly at the highest operating point. However, the same high insulation offers only 38% availability at all operating points in summer. This is due to the energy loss to a fan, as it needs higher airflow to dissipate heat as the operating point increases. The choice of design for 100% availability is straightforward if the operator does not care about the operating point or energy-efficiency. Figure 12 illustrates that all the design options offer 100% availability across all seasons. They only differ by the highest operating point at which they offer

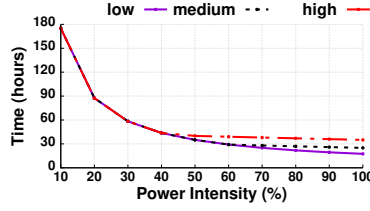


Fig. 13. Performance during summer for high (winter-optimal), medium (spring-optimal), and low (summer-optimal) insulation designs over various operating points.

100% availability. Therefore, if the operator wants the system to be available all the times, they can choose any design and then operate it at the highest operating point at which it offers 100% availability.

Key point. Any design can achieve 100% availability across all the seasons if operated at 30% or lower power intensity. A system-specific design offers the maximum availability at all operating points. **Availability vs. Operating Point.** Each operating point offers different availability guarantees across seasons. For example, at 30% or lower power intensity, the system achieves 100% availability irrespective of the design. This is less than the 50% power intensity that an ideal system can support with 100% availability. It shows the poor thermal management of designs for the non-optimal seasons. It is interesting to note that each design gives 100% availability at a higher operating point in its optimal season. For example, a high insulation manages thermal energy the best in winter, as it exceeds the 50% operating point. This is due to the retention of heat that takes the battery temperature above 25°C and extracts more energy than the nominal value. This effect is consistent across seasons: the design that best manages the thermal energy in a given season, achieves the highest operating point for 100% availability.

Key point. The operating point dictates the availability that system achieves across seasons. The high insulation offers the highest operating point (40%) for the 100% availability across seasons. On the other hand, medium insulation offers the highest availability (45%).

4.1.3 Performance. Figure 13 shows the performance of different designs during the summer. There are two key points in this evaluation. First, at different operating points, the speed of work differs. This is intuitive as power intensity on the x-axis is directly proportional to the CPU utilization. At 100% utilization, the rate of computation is 10 times faster than at 10% utilization. The second takeaway is the difference in performance across designs. The low insulation performs the best as it does not need a fan or active cooling. However, the other two designs must dissipate energy using a fan, as well as use workload scheduling to reduce the energy consumption of the fan. For example, the high insulation design uses a very simple scheduling policy to stop computation when the temperature exceeds 60°C and resumes it only when the temperature drops to 60°C using a combination of passive cooling through conduction and active cooling using the fan. This way, the design is able to achieve higher energy-efficiency at the cost of performance.

4.2 Case Studies

In this section, we present two applications as case studies that prioritize different metrics. The first application is of precision agriculture where an IoT base station is trying to gather sensor data from environmental sensors. The state of the art in this case is the Farmbeats platform [26]. The second application is of federated learning case in smart cities where multiple edge computing platforms are training a machine learning (ML) model. Our evaluation shows that *Acclimator*'s design and operation of these environmentally-powered systems achieves better performance for the systems than the state-of-the-art for each application.

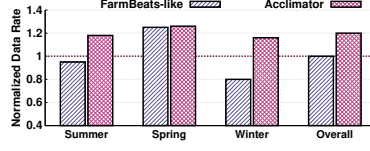


Fig. 14. Performance comparison against Farmbeats. *Acclimator* outperforms Farmbeats thermal-agnostic design during the summer and winter months. Both have the same performance during spring season.

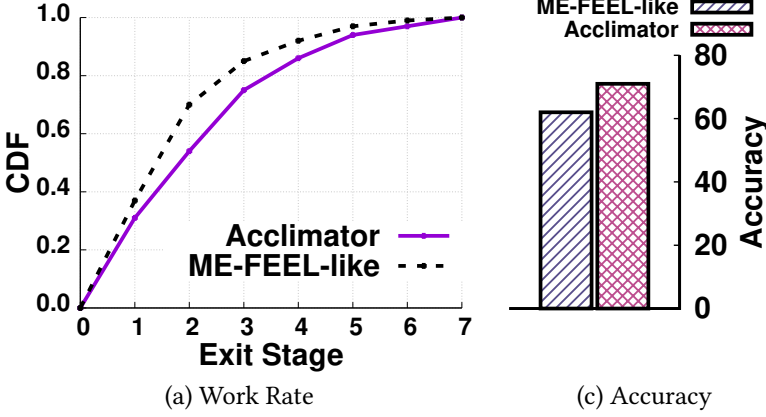


Fig. 15. Performance comparison against a modified MultiExit Federated Edge Learning (ME-FEEL) approach [24]. *Acclimator* outperforms ME-FEEL by exiting in a higher stage leading to a higher accuracy of the trained model.

4.2.1 Sensor Data Acquisition for Precision Agriculture. We evaluate the performance of our system against Farmbeats [26]. The Farmbeats system attempts to minimize the data gaps by varying the duty cycle of the data acquisition. Vasisht et al.[26] details the hardware configurations of the system, but does not provide any details about the system enclosure. The system is powered by two solar panels of 60W each. Solar panels are connected to four 12V-44Ah batteries connected in parallel. The processing component is a Raspberry Pi 4B that consumes 2.7W in idle and roughly 7W at maximum. The environmental sensors are interfaced with the base station through a 802.11b router, that consumes 20W power at maximum with a base power of 3W. We assume a linear relationship between the router power and sensor data acquisition rate.

Figure 14 shows the data rate achieved using a thermal-aware *Acclimator* design and a thermal-agnostic Farmbeats design. Both systems perform similarly during the spring season when the temperatures are moderate and there are no significant variations in temperature over the course of the day. However, the *Acclimator* design outperforms the Farmbeats during summer and winter. The difference is significant during the winter when thermal-aware design is the most important. Overall, the *Acclimator* design achieves 24% higher data rate on average than the Farmbeats default design.

4.2.2 Federated Learning at Resource-constrained Edge. There is significant prior work that focuses on leveraging federated learning on resource-constrained edge devices [24, 25]. These approaches aim to maximize the accuracy of the trained model under energy availability constraints at each device and constraints on the bandwidth available to upload the local model to the edge server. While these approaches consider constraints on the resources, they do not explicitly consider how these constrained resources might be further impacted when exposed to a wide range of extreme environments. This case study shows that by managing thermal energy, in addition to

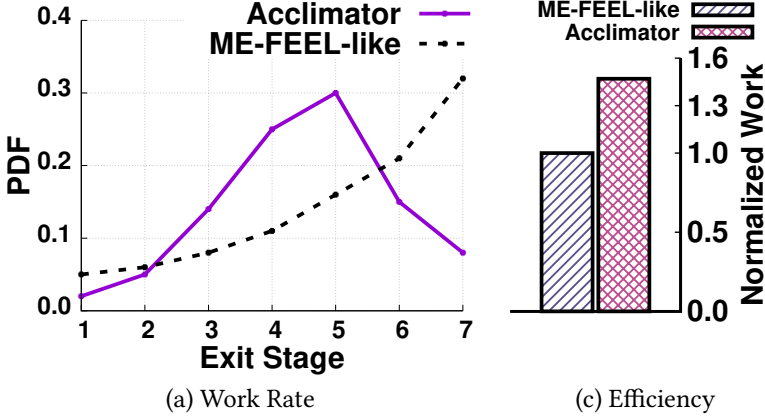


Fig. 16. Energy-efficiency comparison against a modified MultiExit Federated Edge Learning (ME-FEEL) approach [24]. *Acclimator* outperforms ME-FEEL by on energy-efficiency while compromising on the performance.

electrical energy, these systems can achieve better performance, both in terms of energy efficiency and work rate.

Application Setup. Our baseline for this case study is the Multi-Exit Federated Edge Learning (ME-FEEL) approach proposed by Tang et al. [25]. This application setup uses a modified ResNet-18 [14] deep learning model that enables exiting at any of its seven layers. Tang et al. [25] profile the training time for each exit stage that we use in our simulation. Our application scenario consists of periodic rounds where the edge device trains its model and uploads the results to the server within a certain time threshold. The edge device picks an exit stage based on the energy availability at the device. We evaluate two scenarios where an application maximizes its work rate and energy-efficiency over the course of a long period.

Optimizing Work Rate. As illustrated in [25], exiting the training at a higher stage leads to a higher model accuracy. The goal for this application use case is to exit at the highest stage possible in each round of training. To achieve this goal, the application must prioritize the work rate over system availability and energy-efficiency. Figure 15 compares the performance of the ME-FEEL’s thermal-agnostic approach to *Acclimator*’s thermal-aware approach. We can see that the *Acclimator* approach results in a higher probability of exiting at a higher stage. As a result, the model trained with *Acclimator* has a higher accuracy. However, the caveat is that, in an attempt to exit at a higher stage and increase the work rate, the training node may not be able to participate in some of the rounds, since it runs out of energy due to lower energy-efficiency. As a result, while the trained model will have higher accuracy, it may not be trained on the latest data at all times. Our approach achieves an accuracy of 71% as compared to 62% of the baseline, representing a 14.5% improvement in accuracy.

Optimizing Energy-efficiency. The use case above prioritizes model accuracy over model freshness. However, if an application prefers to train the model on the most recent data, it may decide to compromise on its accuracy. To do so, the application will exit at an earlier stage to ensure that it can participate in the next round of training with fresh data. Figure 16 illustrates that *Acclimator*’s approach that prioritizes energy-efficiency and availability over work rate exits at earlier stages more often. However, in doing so, it is able to train more often on recent data leading to a higher amount of work done. *Acclimator* is able to train on more data as quantified by the amount of work done; specifically, it trains on 41% more data than the default thermal-agnostic approach.

5 RELATED WORK

Our work is related to existing work on energy-harvesting sensor systems, sustainable clouds, and data center cooling.

Energy-harvesting Sensor Systems. There has been substantial prior work on designing environmentally-powered systems that dynamically adapt their energy usage to enable perpetual operation, mostly for small-scale energy-harvesting sensor systems [16, 23, 27]. Prior work in this domain can be categorized into two broad categories. First, most of the prior work assumes ideal operating conditions, e.g., 20–25°C and ignores thermal effects. Second, there is prior work that considers battery discharge characteristics and selective thermal effects on the energy-efficiency of the system [3, 8, 10, 21]. The closest work to ours is [21], as it considers the effect of ambient temperature and discharge current on the energy-efficiency of the battery. However, this work does not discuss all the effects considered in our work, e.g., insulation-fan effect, scheduling-energy effect. Also, the evaluation focuses on above 0°C temperatures where some of the effects do not exhibit themselves, e.g., temperature-charging effect.

Overall, to the best of our knowledge, prior work in sensor systems does not discuss enclosure design and how it affects the ability to scavenge and retain thermal energy across different seasons. In addition, no prior work considers all the different effects that impact the energy-efficiency, availability, and performance. Finally, the approaches that propose adapting operation consider only the instantaneous impact on energy-efficiency and not the long-term effects. This is because prior work does not consider or leverage the feedback loop that exists between the system's current operating point and the future available energy capacity.

Sustainable Clouds. There is also recent work on designing sustainable clouds powered by renewable energy with battery storage [2, 11, 12, 18, 22]. This work primarily focuses on adapting the workload to match variations in the energy supply. For example, it uses batteries to either smooth out the fluctuations in power or keep the servers on at minimum capacity to avoid the need for checkpointing or re-computation before resuming operation on the next day when renewable energy becomes available. These small-scale edge data centers would not have the sophisticated cooling and heating systems to provide the ideal operating conditions, as is the case for large-scale data centers. As these small-scale data centers are deployed across different geographical locations, they will be exposed to environments that impact their availability, energy-efficiency, and performance. However, prior work on sustainable clouds has ignored the thermal effects we focus on.

Data Center Cooling. There is significant work on managing heat in large-scale data centers. However, this work focuses largely on the movement of heat within facilities and avoiding hotspots. There is also work on using free cooling for data centers, which discusses the impact of ambient temperature on energy-efficiency and how the workload should be adapted to optimize cooling efficiency [9, 13]. However, the thermal effects in free-cooled data centers are a lot less restricting than the constraints imposed on environmentally-powered systems. Specifically, large-scale data centers are not powered by batteries and thus generally do not discuss heat management for batteries [20]. Data centers also focus on cooling only; the heat produced is considered waste that must always be dissipated. In contrast, environmentally-powered systems can productively leverage this heat in cold weather.

6 CONCLUSION

Our results demonstrate the complexity and importance of designing and operating environmentally-powered computing systems. These systems must consider, not only their electrical energy, but also the thermal energy available in the environment in both their design and

operation. Our work takes an important step in that direction by enumerating, quantifying, and modeling these systems' thermal effects using a small-scale prototype and programmable incubator. We then leverage our thermodynamic model to show how considering these thermal effects in designing and operating environmentally-powered computer systems can improve their energy-efficiency, performance, and availability. Our evaluation shows that a season-specific design can achieve up to 35% higher energy-efficiency than a non-optimal design. It also outperforms the non-optimal design by achieving 20% higher availability. Similarly, the right choice of operating point allows achieving up to 20% higher energy efficiency. Finally, our case studies show that the thermal-aware design of *Acclimator* yields an improvement of 24% in data acquisition rate for precision agriculture application, 14% increase in model accuracy for the federated learning at the edge, and 41% increase in the data used for training at the edge.

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